

Technical Datasheet - VBF-T6R1NOCEILING-DSH-01

ED-Compact B-p smartphone cell. Values mechanically taken from parameter set / simulation outputs and annotated line by line; missing items honestly stated.

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Field	Value (with source)
Product / electrochemical system	ED-Compact B-p smartphone cell: NMC811 positive / graphite negative (Chen2020 parameter set). Electrolyte: EC:EMC + LiPF6 (parameter-set formulation; transport: Nyman2008 functions); cation transference number 0.2594.
Rated capacity (Ah)	5.6 Ah nominal (design value, loading-reconciled in params_r2_bp.json); 1C simulation-verified 5.3222 Ah (cell/r2_bp_1c.json:capacity_ah).
Nominal voltage / voltage window (V)	Discharge-time midpoint 3.9214 V (cell/r2_bp_energy.json:midpoint_voltage_v); window 2.5 - 4.2 V (parameter set cut-offs).
Rated energy (Wh)	18.6012 Wh (1C discharge integration, cell/r2_bp_energy.json:energy_wh).
Energy density (Wh/L / Wh/kg)	964.44 Wh/L; 469.06 Wh/kg (contract caliber: electrodes + current collectors + separator; electrolyte excluded - calc-energy definition).
Maximum continuous discharge rate	1C (5.6 A) simulated (SPMe, 25 C). Higher rates not verified - honest.
Fast-charge capability	4C NOT supported: at 4C charge from empty, 45 C ambient (DFN, lumped thermal, h = 10 W/m2K) the anode potential minimum reaches -0.2566 V -> lithium plating (cell/r2_bp_4c.json:anode_potential_v), T_max 351.0 K (same file), 4C charge acceptance 0.096 Ah before the 4.2 V limit. Design-side recommendation: derate charge to <= 2C (qualitative, not simulation-verified).
Operating temperature range	Simulated at 25 C ambient (1C discharge, 1C aging) and 45 C ambient (4C charge). T_max at 4C/45 C exceeds the 50 C limit -> 4C operation not permitted. Honest operating statement: <= 45 C ambient with derated charge.
Cycle life	Aging simulated (aging_1c_100cyc, isothermal 25 C): anode SEI 417.74 nm after 100 cycles - within the 500 nm limit (cell/r4_bp_aging.json:sei_thickness_nm_end). Capacity trajectory 1.798 -> 2.151 Ah shows the climb-then-saturate artifact (lithium-loss window shift; Chen2020 set starts discharged) - annotated, NOT treated as normal degradation. Cycle life to 80% capacity: Not simulated (beyond the aging protocol).
Safety determination	4C plating: FAIL (plated). 4C/45 C temperature: FAIL (351.0 K > 323.15 K). Anode SEI after 100 cycles: PASS (417.74 nm <= 500 nm). Mechanical verdicts from log-evaluate (batch r4).
Dimensions and mass	Electrode strip 1580 mm x 65 mm; stack thickness 187.8 um (pos 75.6 / neg 85.2 / sep 9 / Al 10 / Cu 8 um); contract volume 19.29 cm3; mass 39.66 g (contract caliber, electrolyte excluded) + electrolyte 5.42 g (pore volume 4.514 cm3 x 1.2 g/cm3 literature density).
DC resistance	0.364 mohm (10%-discharge definition, cell/r2_bp_energy.json:dcr_ohm).
Power density	283313 W/kg (DCR-based, cell/r2_bp_energy.json:power_density_w_kg).
Voltage plateau vs spec	3.9214 V vs >= 4.1 V criterion -> FAIL (NMC811 cathode OCP ceiling ~4.0 V; recorded in log final.escalation).